$\omega = \frac{}{(2\pi f)} \quad (2\pi(1.5))$

18 Ans: B

For a point source, intensity $I \propto 1/r^2$ (1)
For all waves, intensity $I \propto (\text{amp, } A)^2$ (2)
Thus, amplitude $A \propto 1/r$
Substituting, $8X \propto 1/r$ (A)
Final amplitude $\propto 1/(2r)$ (B)
(B)/(A): Final amplitude = 4X (B)

19 Ans: D